Faculty of Media Engineering and Technology
Dept. of Computer Science and Engineering
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CSEN 605: Digital System Design
Winter2025

Practice assignment 6-2

VHDL Testbenches in Quartus

Step 1: Create a project and create the following code. (same settings as previous tutorial)

```
library ieee;
use ieee.std_logic_1164.all;
entity project1 is
port( A,B,C: in std_logic; F: out std_logic);
end project1;
architecture arch of project1 is
begin
F <= (NOT A and NOT B) OR C;
end arch;
```

Step 2

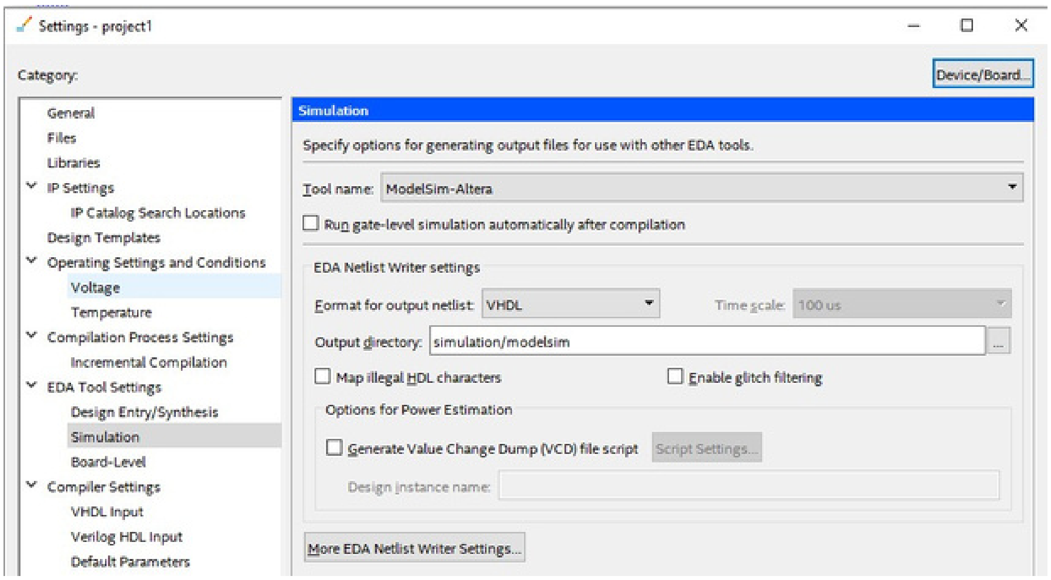
We want to create a TestBench to simulate this codebyassigning values to inputs that change according to some planned time and see how the output is behaving.

To do this, we must create a testbench and simulate in modelsim later.

2.1) - Go to assignments settings EDA tool settings simulation: Choose modelsim altera for tool name and VHDL for netlist.

Note that the output directory is a folder (and subfolder) called simulation/modelsim. Keep them.

- Click ok and exit.

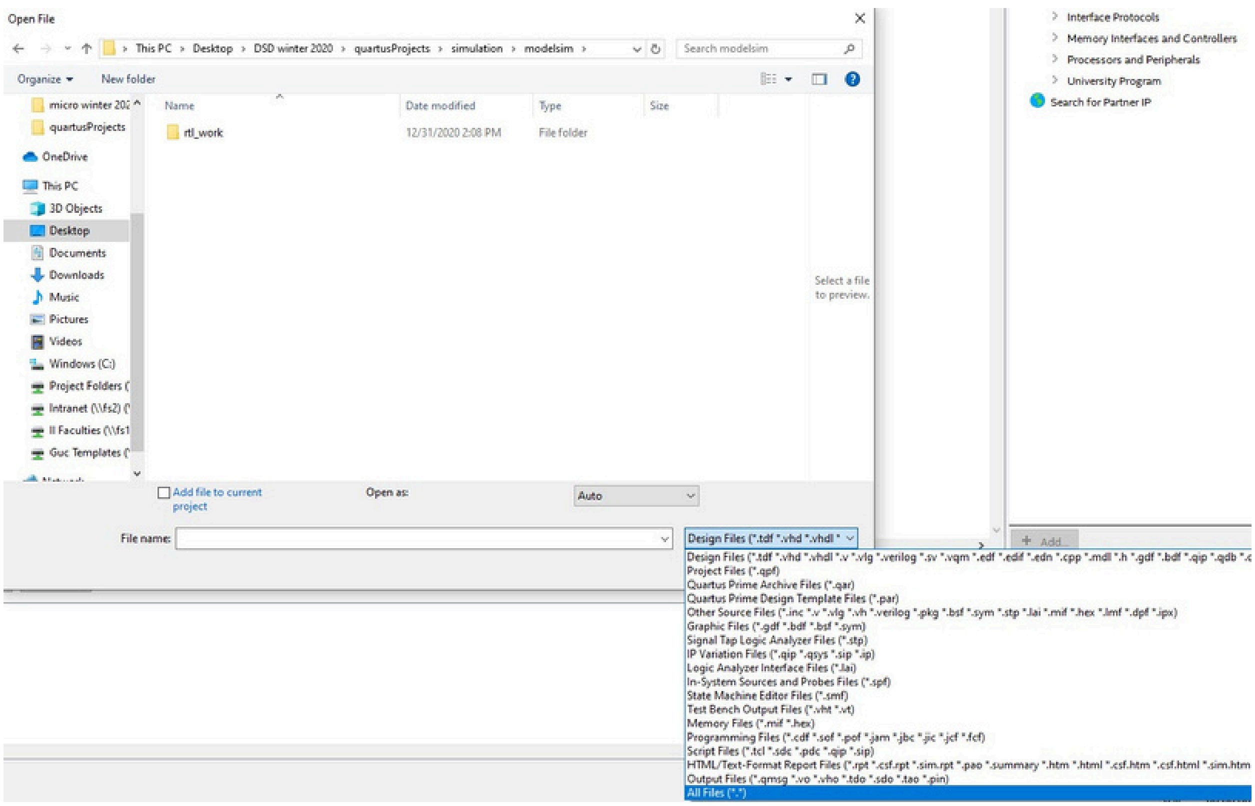


2.2) To create your test bench, you must go to
Processing → Start → Start test bench template writer.

This will create a testbench file (already filled with code) called [**projectname].vht**

To open this file to view its contents, just go **to file open**, and then navigate to your project folder, go inside the simulation folder, then modelsim folder and you should find it.

*****N.B. Make sure to select "all files" from the file filter in the bottom right to show all file types. Otherwise, your vht file will not show.**

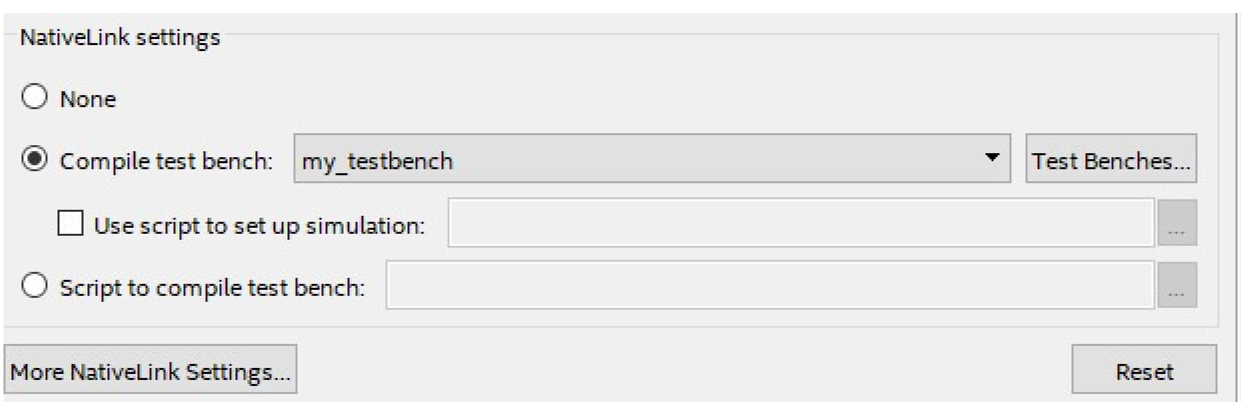


2.3) The testbench is shown. Make sure to edit it to look like this. We added 3 processes for a, b and c. We are setting them to values and waiting for few ns then changing the values. Remember we are simulating later.

```
LIBRARY ieee;

USE ieee.std_logic_1164.all;


ENTITY  project1_vhd_tst  IS  END  project1_vhd_tst;

ARCHITECTURE  project1_arch  OF  project1_vhd_tst  IS

SIGNAL A : STD_LOGIC; SIGNAL B : STD_LOGIC; SIGNAL

C  :  STD_LOGIC;  SIGNAL  F  :  STD_LOGIC;  COMPONENT

project1




      PORT (A : IN STD_LOGIC; B : IN STD_LOGIC; C : IN STD_LOGIC; F : BUFFER
STD_LOGIC);
END COMPONENT;

BEGIN

      i1 : project1

      PORT MAP (A => A, B => B, C => C, F => F);


process_A: PROCESS

BEGIN

             A<='0';

             wait for 100 ns;

             A <='1';

             wait;


END PROCESS process_A;
```

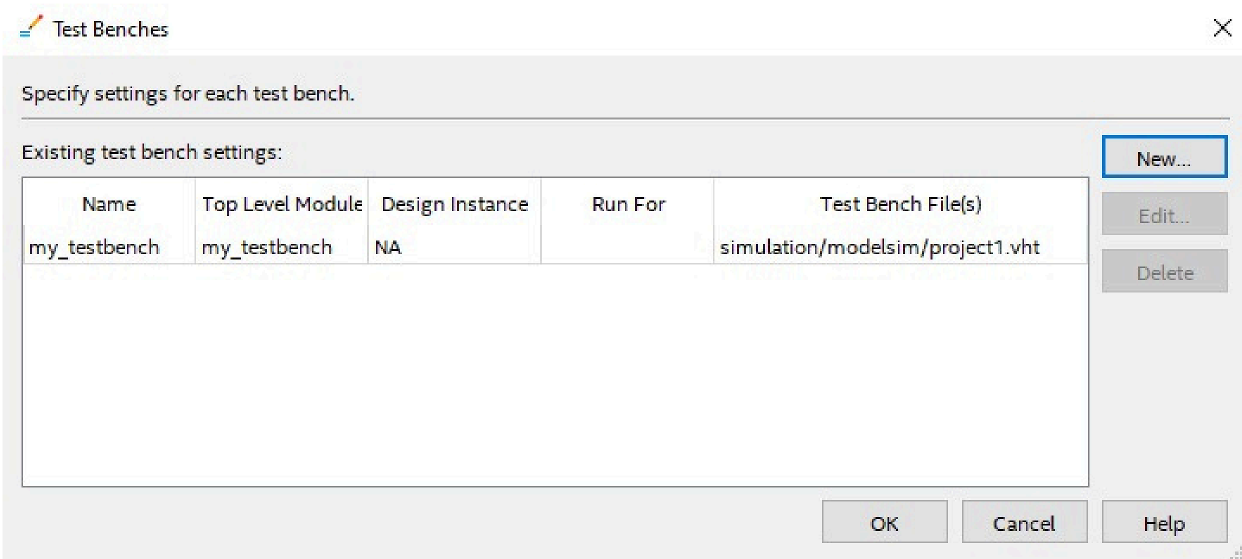
```
process_B: PROCESS
BEGIN

             B<='0';
             wait for 150 ns;
             B <='1';
            wait;

END PROCESS process_B;

process_C: PROCESS
BEGIN
             C<='1';
             wait for 175 ns;
             C <='0';
            wait;

END PROCESS process_C;
END project1_arch;
```

2.4) To compile the testbench, go to **assignments** **settings** **EDA tool settings** **simulation** and under nativelink settings, select compile test bench then click on test benches button on the right.



(initially, the field containing my_testbench is empty)

After you click on test benches button, you get this:



Click on new, choose a name for your testbench (my_testbench for example), then click on the 3 dots next to file name, browse to your testbench file and click on add.

You should get something similar to the previous image.

Click ok and ok and exit.

2.4) click on project → add current file to project (just to make sure your testbench is added to your project)

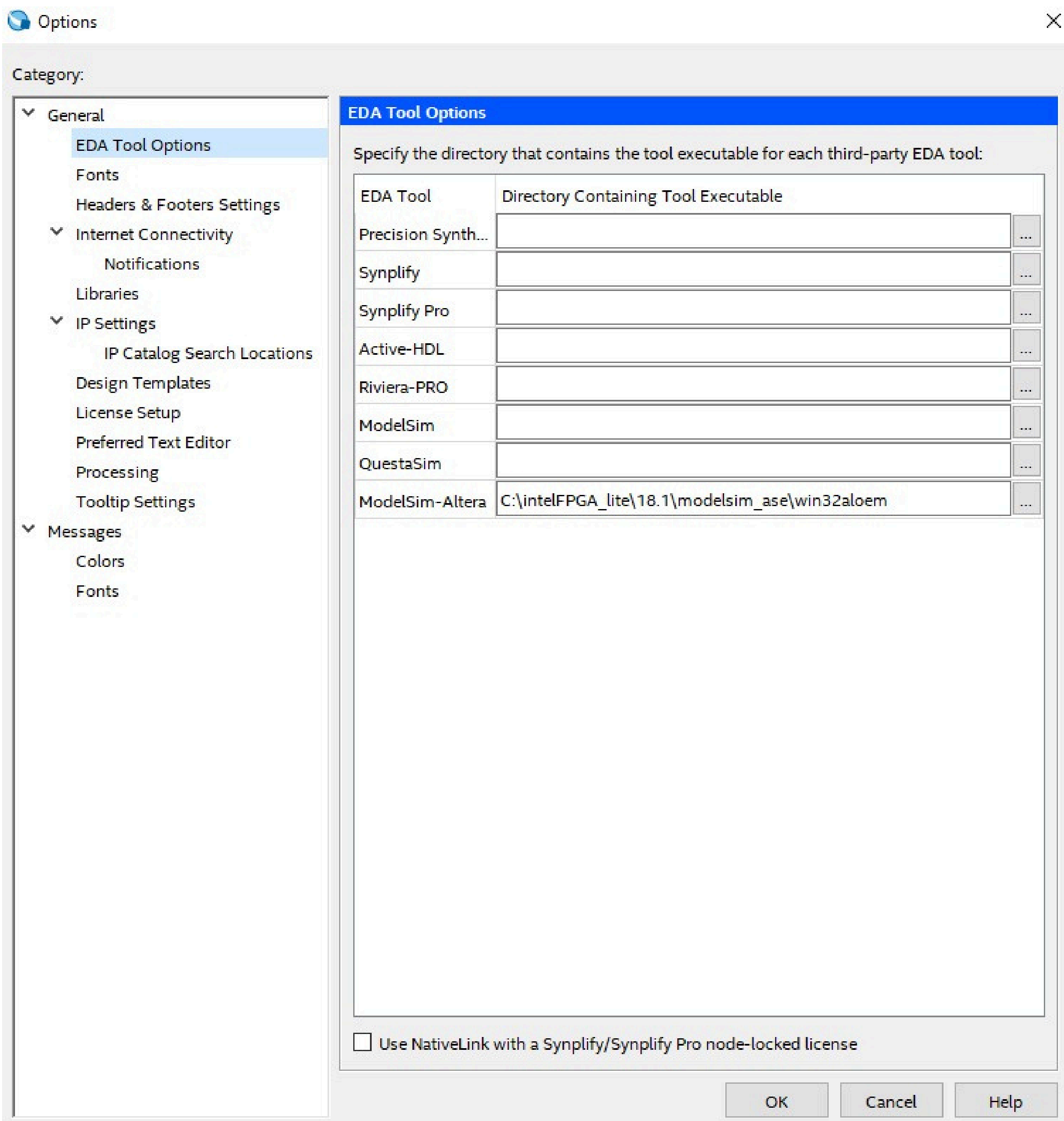
2.5) compile (processing → start compilation)

2.6) to simulate in modelsim, we must tell quartus that.

Go to tools → options→ EDA tool options

Under modelsim-altera (not modelsim), choose the correct directory. It should be found in [yourInstalledQuartusLocation]\modelsim_ase\win32aloem

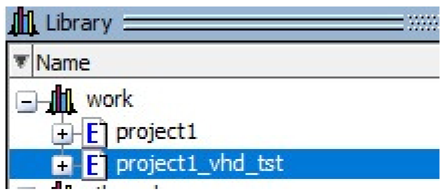
Example on my computer: C:\intelFPGA_lite\18.1\modelsim_ase\win32aloem



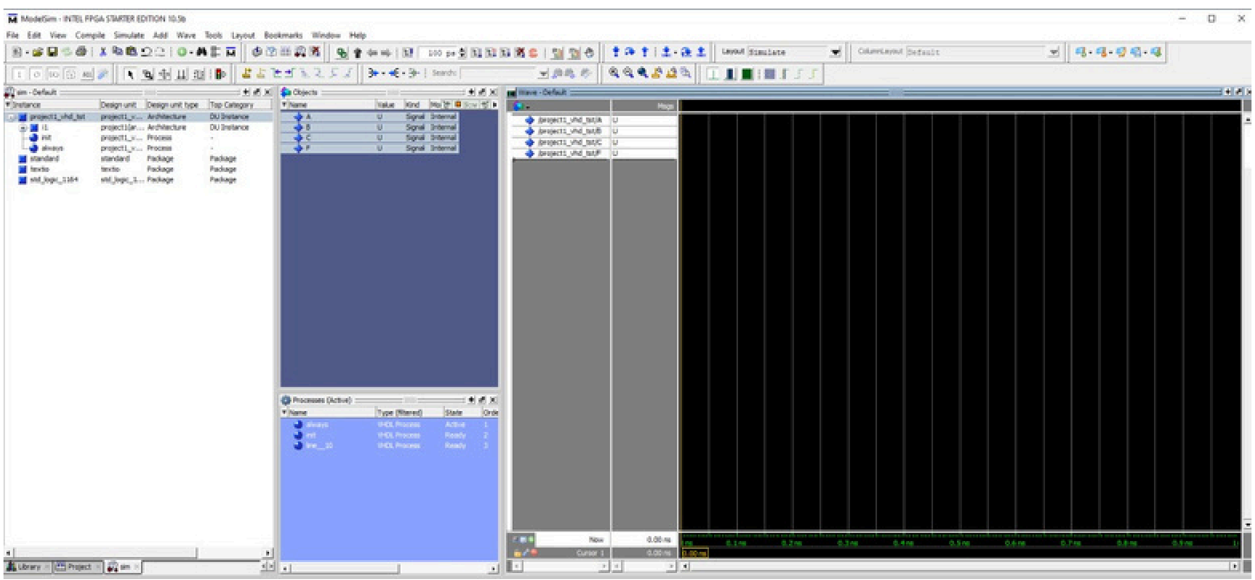
2.7) Simulation: **go to tools Run simulation tools** □**RTL simulation**

This should open modelsim.

In Modelsim, under the work library, expand your project, and locate the file ending with vhd_tst. This is your testbench file.



Double click on it (or right click simulate), and select all the signals and drag them to the wave.



Change the step size to 100ns and run few steps